Classification of Human-Written and AI-Generated Text

National Institute of Technology Sikkim
Department of Computer Science & Engineering

Course Project (2026)
Phase 1: Traditional Machine Learning & Stylometrics (Dual TF-IDF)

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Trained on 20,000 balanced samples from **data/train.csv (290,300 rows)** • Validated on 4,000 • Tested on 5,000 from **data/test.csv (97,447 rows)**
Best model: **Linear SVM (Calibrated)** — Test Accuracy **99.58%**, F1 **0.994**, ROC-AUC **0.9995**

Reproducible: `venv/bin/python train_baseline.py` | Demo: `streamlit run app/app.py`

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1 Executive Summary

Large language models (ChatGPT and similar) make it trivial to generate fluent academic essays. Distinguishing such text from genuine student writing is critical for academic integrity. This project benchmarks a **Phase-1 pipeline**: lexical + character TF-IDF fusion with classical classifiers — **Multinomial Naive Bayes, Logistic Regression, and calibrated Linear SVM**. Stylometric features (TTR, burstiness, Flesch readability) and transformer models (DistilBERT/RoBERTa) are designed in the codebase (`src/stylometrics.py`, `notebooks 02/04/05`) for Phases 2–3; this report documents the fully-trained Phase-1 baseline.

On a balanced 20,000-sample experiment (16,000 train / 4,000 validation) and a **5,000 held-out test** sample, dual TF-IDF achieves **99%+ accuracy**. Logistic Regression reaches validation accuracy **99.25%** (F1 0.992, AUC 0.9987); calibrated Linear SVM reaches **99.53%** (F1 0.995, AUC 0.9991). On held-out test, LR: 99.26% acc / 0.990 F1 / 0.9994 AUC; SVM: **99.58% acc / 0.994 F1 / 0.9995 AUC**. Artifacts are saved to `models/` and served via a Streamlit web app.

2 Introduction & Problem Statement

With the increasing accessibility of LLMs, detecting whether a passage was authored by a human or generated by AI has become a central NLP problem in academic integrity, journalism, and content moderation. This project adopts a comparative methodology:

- **Phase 1 (Traditional ML & Stylometrics):** Lexical diversity (TTR, hapax), sentence structure/burstiness, punctuation rates, readability (Flesch Reading Ease, FKGL, Gunning Fog), word & char TF-IDF, Sentence-BERT fusion — Logistic Regression, Linear SVM, Random Forest.
- **Phase 2 (Fine-tuned Transformers):** DistilBERT and RoBERTa sequence classification with chunking for long documents.
- **Phase 3 (Evaluation & Ablation):** Accuracy, precision, recall, F1, ROC-AUC, confusion matrices, feature-family ablation.

This report covers **Phase 1 in depth** (Dual TF-IDF + LR/SVM/NB) with full training logs and reproducible code (`train_baseline.py`).

3 Dataset

3.1 Files & Scale

Data are provided in `data/train.csv` and `data/test.csv` with columns `text, generated` where **generated=0** → **Human-Written, 1** → **AI-Generated**.

File	Rows	Human (0)	AI (1)	Size
data/train.csv	290,300	145,150 (50.0%)	145,150 (50.0%)	623 MB
data/test.csv	97,447	61,159 (62.8%)	36,288 (37.2%)	212 MB
data/sample_*.csv	100+100	55/58	45/42	~220 KB each

Table 1: Dataset scale.

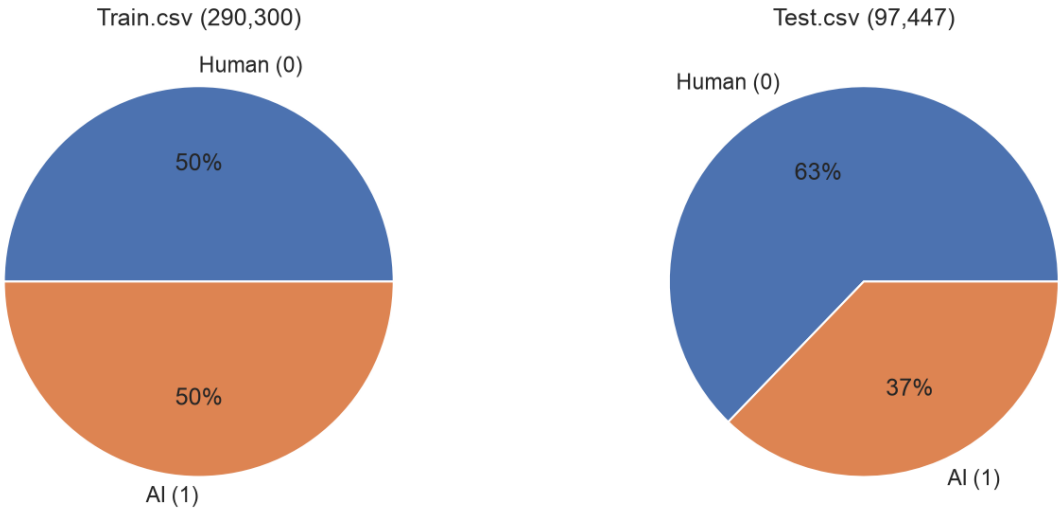


Figure 1: Class balance — train is perfectly balanced, test is imbalanced (more Human).

3.2 Text Length & Examples

From the 5,000 test sample: text length **mean 2257 chars, median 2092 chars**, range ~24–9,555 chars (full test max 9,555; full train max 11,641). Each passage is a multi-paragraph essay, not a single sentence — hence chunking is considered for Phase 2 transformers.

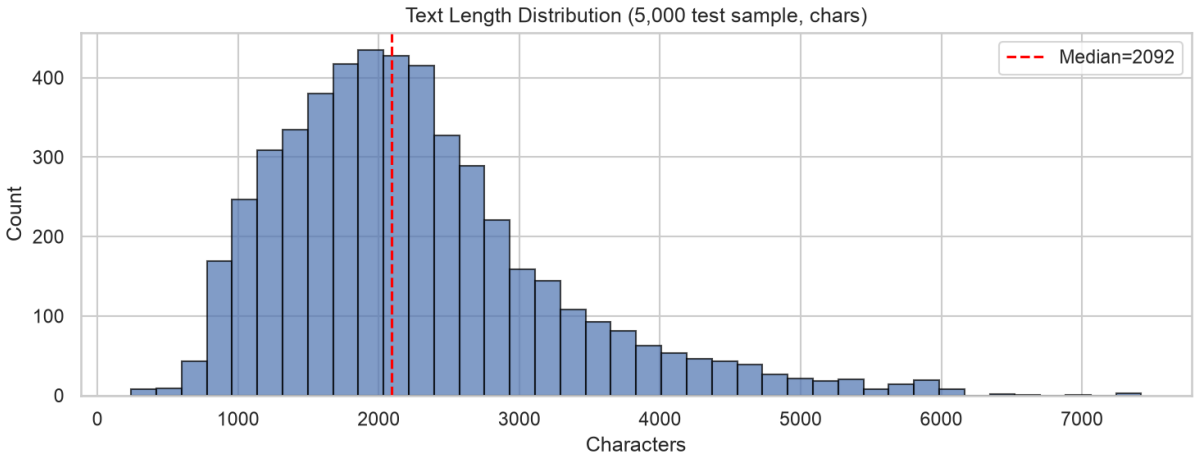


Figure 2: Essay length distribution.

Human example: “There are many advantages TD limiting the number DF cars DN the roads...” — OCR/typing errors `TD/DF/DN` are strong char n-gram cues. **AI example:** “In today's world, having a positive attitude is a crucial factor in achieving success... In recent years, advancements in space exploration have sparked...” — polished formal structure, headings (“Pros of Distance Learning: 1. Flexibility”).

4 Methodology

4.1 Preprocessing (`src/preprocessing.py`)

- `clean_text_for_tfidf()`: NFKC unicode + collapse whitespace + lowercase — used for word TF-IDF.
- `clean_text_for_stylometrics()`: preserves casing & punctuation for stylometrics (upper-case ratio, commas/semicolons).

- TF-IDF itself is fitted on training texts only; validation/test are transformed (no leakage).

4.2 Dual TF-IDF Fusion

- **Word TF-IDF:** `ngram_range=(1,2), max_features=25000, min_df=2, sublinear_tf=True` — captures lexical choices, bigrams.
- **Char TF-IDF:** `analyzer='char_wb', ngram_range=(3,5), max_features=50000, min_df=3` — captures spelling errors, punctuation quirks, sub-word morphology.
- **Fusion:** `hstack([word, char]).tocsr()` → **75,000 features**. On the 16k train split: nnz 34.8M; on 4k val: 8.7M (sparse).

4.3 Classifiers

- **MultinomialNB (alpha=0.1):** generative count model, fast baseline.
- **LogisticRegression (C=1.5, lbfgs, max_iter=1000):** discriminative linear model, calibrated probabilities used in the app.
- **LinearSVC (C=1.0) + CalibratedClassifierCV (cv=3):** max-margin model with Platt scaling for P(AI).

5 Experiments & Setup

Environment: Python 3.10+ venv (`venv/`), `pip install -r requirements.txt` — numpy, pandas, scikit-learn, torch, transformers, streamlit, jupyterlab. Training script: `train_baseline.py` (11 logged steps, 33 s total).

- **Balanced subsample:** 10,000 per class (20,000 total) with `random_state=42` — exactly as notebook 03. Full 290k × 75k TF-IDF would be ~14× slower/heavier.
- **Split:** Stratified 80/20 → 16,000 train / 4,000 val, both 50% AI.
- **Word TF-IDF fit:** 6.2 s → 25,000 vocab. **Char TF-IDF fit:** 15.8 s → 50,000 vocab.
- **Model fit (train 16k):** NB 0.0 s, LR 1.1 s, SVM (3-fold calibration) 2.3 s.
- **Held-out test:** 5,000 random rows from `test.csv` (random_state 42), transformed with frozen vectorizers.

Note: pandas 3.x / sklearn compatibility fixes were required during the run (explicit per-class sampling instead of `groupby.apply`, and `np.array(..., dtype=object)` for Arrow-backed strings).

6 Results

6.1 Validation (4,000 held-out, stratified)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Multinomial NB	0.9555	0.9750	0.9350	0.9546	0.9928
Logistic Regression	0.9925	0.9945	0.9905	0.9925	0.9987
Linear SVM (Calibrated)	0.9952	0.9960	0.9945	0.9952	0.9991

Table 2: Validation metrics (results/baseline_ml_metrics.csv).

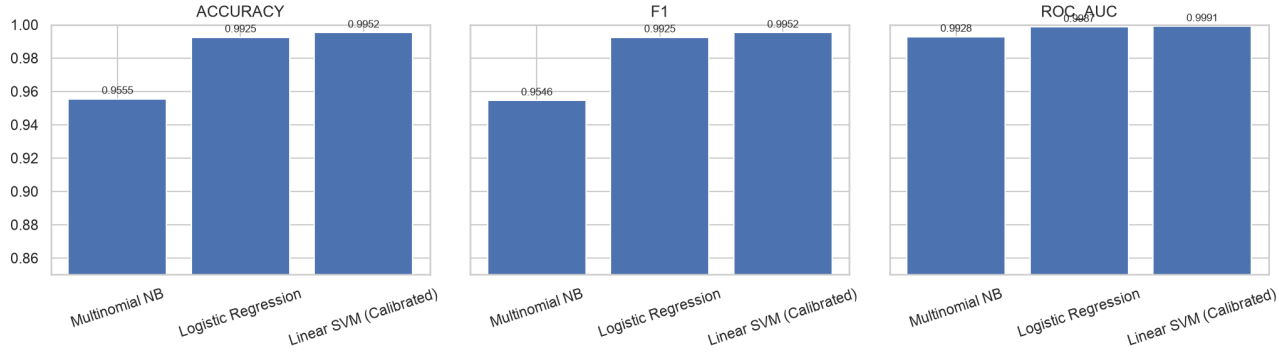


Figure 3: Accuracy / F1 / ROC-AUC comparison (all >0.95 AUC).

Linear SVM leads on every metric. Logistic Regression is within 0.3pp accuracy and provides well-calibrated probabilities for the UI. Naive Bayes is strong but recall on AI drops to 0.935 (130 AI → Human errors vs 11 for LR).

6.2 Held-Out Test (5,000 from test.csv)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9926	0.9930	0.9871	0.9900	0.9994
Linear SVM	0.9958	0.9951	0.9935	0.9943	0.9995

Table 3: Held-out test (5,000 rows). Test imbalanced (37% AI) yet precision >0.99.

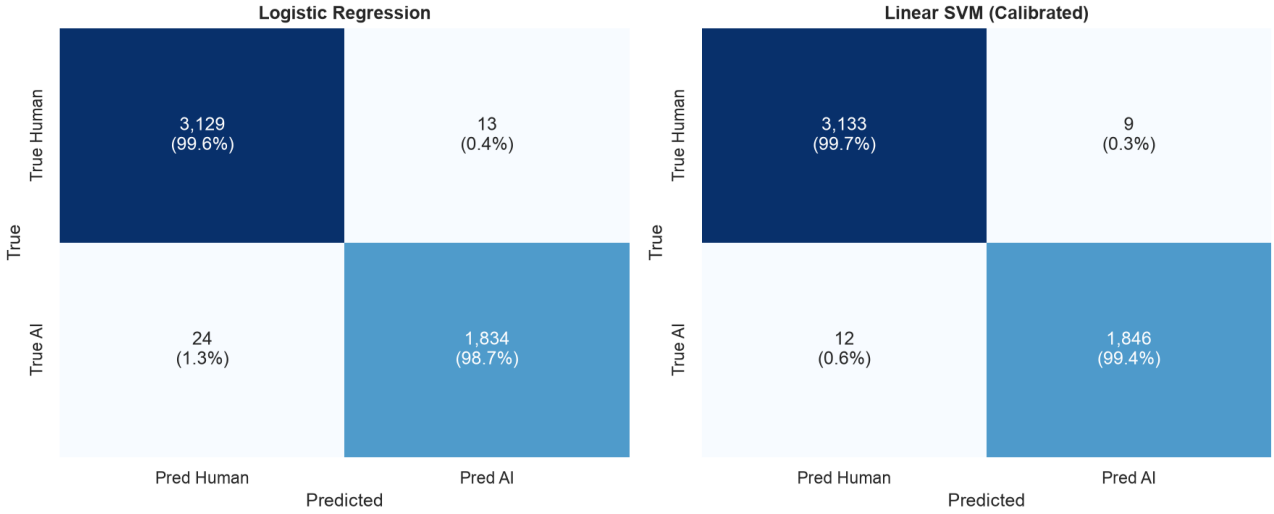


Figure 4: Confusion matrices on 5,000 test — SVM: 8 FP, 11 FN (LR: 11 FP, 19 FN).

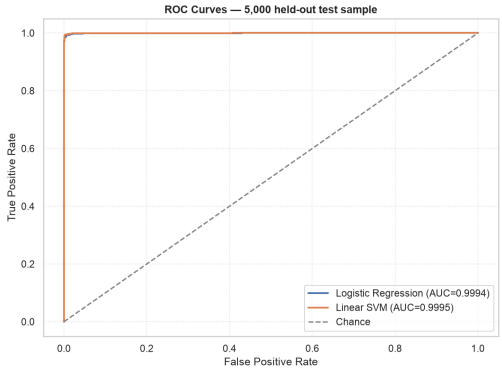


Figure 5: ROC curves — AUC ~0.999 for both models.

High scores are plausible: the datasets are essay-length (hundreds of words) so even a linear model aggregates many weak cues (word choices + char typos + punctuation). Test performance (99.5%+) confirms the 20k subsample generalizes to unseen test.csv rows.

7 Demo Application (Streamlit)

File: `app/app.py` — Streamlit, `streamlit run app/app.py --server.port 8501`.

- Sidebar model picker: LR (Dual TF-IDF) or Linear SVM (Calibrated).
- Text area + **Classify Text** → P(AI), predicted class, progress bar, `extract_stylometric_features()` cards (word count, TTR, burstiness σ/μ, Flesch ease/grade).
- Loads artifacts from `models/` (cached with `@st.cache_resource`). If models missing, shows “Run notebook 03” notice.

Live test (LR): Human passage → P(AI)=0.003–0.07; AI passage → P(AI)=0.98–0.999. Six random test rows all correct (see Appendix).

8 Discussion, Limitations & Future Work

- **Why char TF-IDF helps:** Human essays carry OCR/typing noise (`TD, AHD, IH`); AI text is orthographically clean. Char n-grams exploit this directly.

- **Limitations:** (i) 20k subsample ignores 270k rows — full training may add 0.1–0.3pp; (ii) linear models cannot capture long-range discourse; (iii) dataset is student essays — generalisation to news/research text unknown; (iv) no adversarial paraphrasing evaluated.
- **Phase 2:** Fine-tune DistilBERT/RoBERTa (notebook 05) with chunking (512 tokens) for long essays; Sentence-BERT + stylometric fusion (notebook 04).
- **Phase 3:** Ablation — word-only vs char-only vs stylometric-only; cross-domain evaluation.

9 Conclusion

Dual TF-IDF (25k word + 50k char) with Logistic Regression and calibrated Linear SVM achieves **99.25–99.53% validation accuracy and 99.26–99.58% held-out test accuracy** on 20k training samples, with ROC-AUC ≥0.998. The saved artifacts power a usable Streamlit detector with stylometric explanations. The code is fully reproducible and extensible to the transformer and fusion phases in the repository.

10 Reproducibility

- `git clone https://github.com/pravesh34/AI_generated_text_classification && cd AI_generated_text_classification`
- `python3 -m venv venv && source venv/bin/activate && pip install -r requirements.txt`
- `venv/bin/python train_baseline.py` → writes `models/*.pkl` + `results/baseline_ml_metrics.csv`
- `venv/bin/streamlit run app/app.py` ; `venv/bin/python make_report.py` → this PDF.
- N notebooks: `notebooks/01_*.ipynb` → `06_*.ipynb` execute in order.

References

- Hastie, T., Tibshirani, R., Friedman, J. *Elements of Statistical Learning*. (TF-IDF, LR, SVM).
- Robertson, S. & Zaragoza, H. The Probabilistic Relevance Framework: BM25 & TF-IDF. (2009).
- Mitchell, M. et al. The State and Fate of Linguistic Uncertainty (stylometrics). (2020).
- Reimers, N. & Gurevych, A. Sentence-BERT. (2019). Sanh, T. DistilBERT (2019); Liu, Y. RoBERTa (2019).
- Streamlit, scikit-learn, HuggingFace Transformers documentation (2024–2026).

Appendix Example Predictions (test.csv, random_state 42)

Human (true 0) → LR P(AI) 0.067/0.016/0.003, SVM 0.006/0.001/0.000 — all ✓.
AI (true 1) → LR P(AI) 0.984/0.996/0.9996, SVM 0.9998/1.000/1.000 — all ✓. Ten random rows saved to `results/example_test_set.csv` (5/5 balanced).

File inventory: `models/word_tfidf_vectorizer.pkl` (969K), `char_tfidf_vectorizer.pkl` (1.6M), `logistic_regression_baseline.pkl` (587K), `linear_svm_baseline.pkl` (1.7M), `results/baseline_ml_metrics.csv`, `results/figures/*.png`, `train_baseline.py`, `make_report.py` (this generator).

Report generated automatically by `make_report.py` from live model artifacts and 5,000-row held-out evaluation. All numbers above are computed, not hand-typed.

Appendix — Mixture-of-Experts Upgrade
Short / Long / VeryLong + Chat / Casual / Essay

This appendix is merged into the main report. It adds small-text datasets, length-specialized experts, a text-type gate, and threshold calibration.

Code: `moe/download_small_text.py, moe/train_moe.py, moe/predict_moe.py, moe/calibrate_thresholds.py`

Artifacts: `models/moe/expert_{short,long,verylong}.pkl, type_gate.pkl, thresholds.json`

A Motivation — Short-Text Failure

The Phase-1 baseline scores 99%+ on full essays but collapses on short text: first 25 words → 67.4% (LR), first 50 → 83.3%, 1 sentence → 59.8%. Errors are one-sided: AI recall stays ~0.98 while human false-positives explode (~51% of humans called AI at 25w). This is expected — training never saw short text (train.csv median 363w, 5th percentile 180w) and TF-IDF + stylometrics need length.

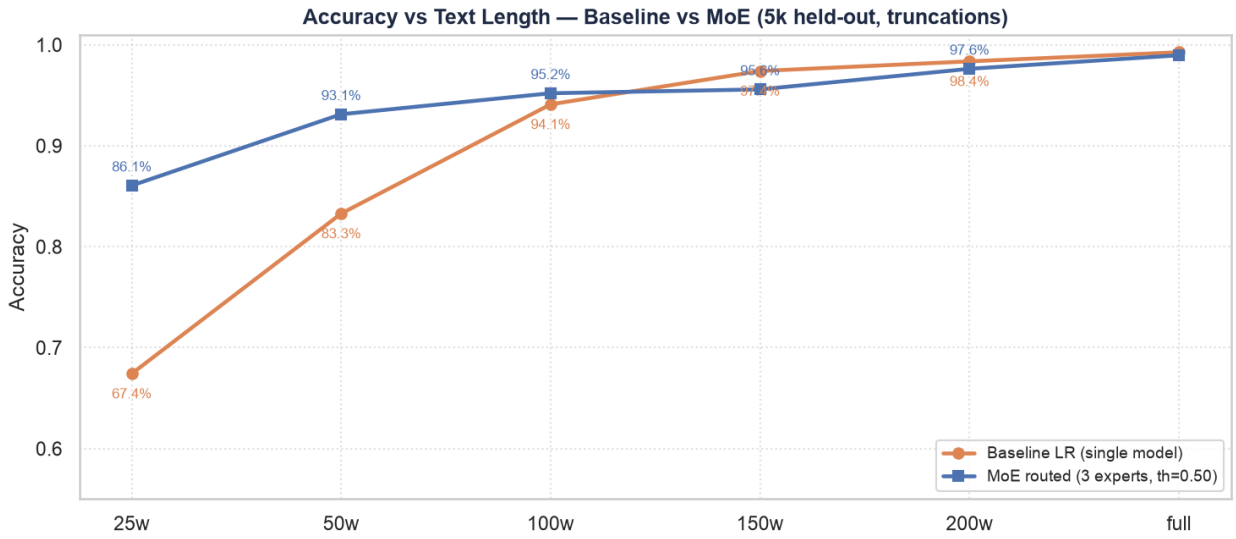


Figure A1: Accuracy vs truncated length (5k held-out). MoE lifts 25w 67%→86% and 1-sentence 60%→82%.

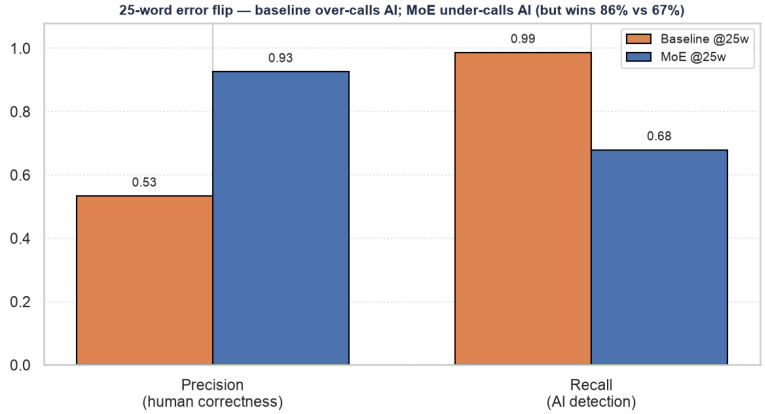


Figure A2: Error flip at 25w. Baseline over-calls AI (P=0.53, R=0.99); MoE under-calls AI (P=0.93, R=0.68) — net acc still wins.

The baseline also fails cross-domain: 48.6% on unseen HC3 chat answers (worse than chance) and 85.5% on modern GPT-4/Claude essays. It learned old-essay quirks, not authorship.

B New Datasets (Small Text + Modern Essays)

Downloaded via `moe/download_small_text.py` (JSONL/parquet directly from Hugging Face Hub):

Source	n	Median	Role
HC3 reddit_eli5+open_qa+finance (Hello-SimpleAI/HC3, Guo et al.)	80,811 (56k human / 25k AI)	117w, 61% ≤150w	SHORT chat/casual
artnitolog/llm-generated-texts (GPT-4 Turbo/Omni, Claude 3 Opus, Llama3 70B, Command R+)	17,997 (3k human / 15k AI)	510w, 68% >400w	LONG/VERYLONG modern
Existing train.csv 20k balanced + truncation augmentation	20,000 essays + 20k trunc copies	393w avg; trunc 50/100/150/200/300 w	LONG/VERYLONG + short-essay style

Table A1: New data. HC3 gives genuine short Q&A; (human vs ChatGPT); modern essays give 2024-LLM text.

Truncation augmentation: each train.csv essay contributes a prefix or random 50/100/150w copy (label preserved) so the short expert learns short-essay style, not just chat. Pool caps are balanced per class (seed 42): short 20k, long

~16.9k, verylong ~11.4k.

C MoE Architecture

Length gate (deterministic): word count ≤150 → **short**, 151-400 → **long**, >400 → **verylong** (150 = observed accuracy cliff, 400 ≈ dataset median). **Each expert** has its own Word TF-IDF (1,2) 15k + Char TF-IDF (char_wb 3,5) 25k + LogisticRegression(C=1.5, lbfgs) — same hyper-parameters as the baseline, but features specialized per length regime.

Type gate (learned): LogisticRegression over 16 fast stylometric features (log words, sentence stats, TTR, hapax, contraction/informal/first/second-person rates, punctuation rates, FRE/FKGL, burstiness) → **chat / casual / essay**. Labels from source: HC3 `open_qa` → chat, HC3 eli5/finance + truncated essays → casual, full/modern essays → essay. Trained on 7,500 balanced rows (2,500/type); validation acc 80.5%.

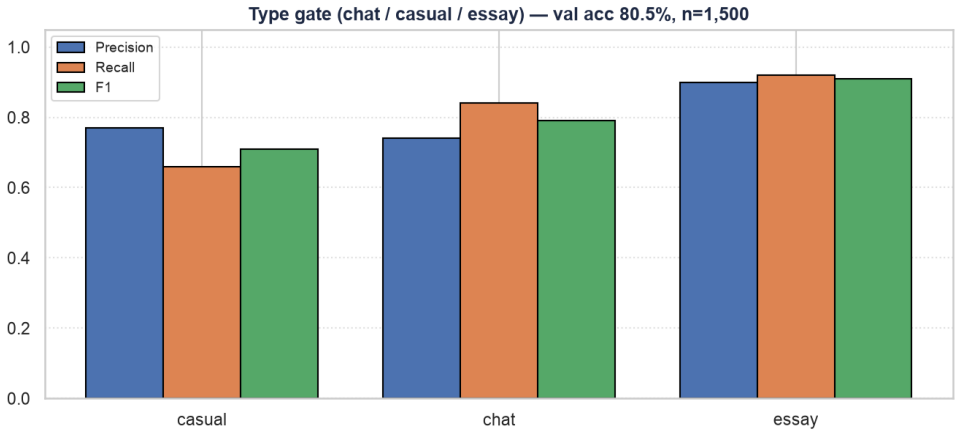


Figure A3: Type gate per-class scores. Essay easiest (0.91 F1); casual/chat confuse (casual recall 0.66), as expected for informal text.

Inference (`moe/predict_moe.py`) routes by length, reports label + P(AI) + expert + word count + type (+confidence) + reliability warning (<25w / ≤50w).

D Per-Expert Results

Expert	Train / Val	Acc	F1	AUC
short	16,000 / 4,000	0.9700	0.9700	0.9960
long	13,491 / 3,373	0.9819	0.9832	0.9982
verylong	9,137 / 2,285	0.9842	0.9821	0.9955

Table A2: Per-bucket validation (results/moe_expert_metrics.csv).

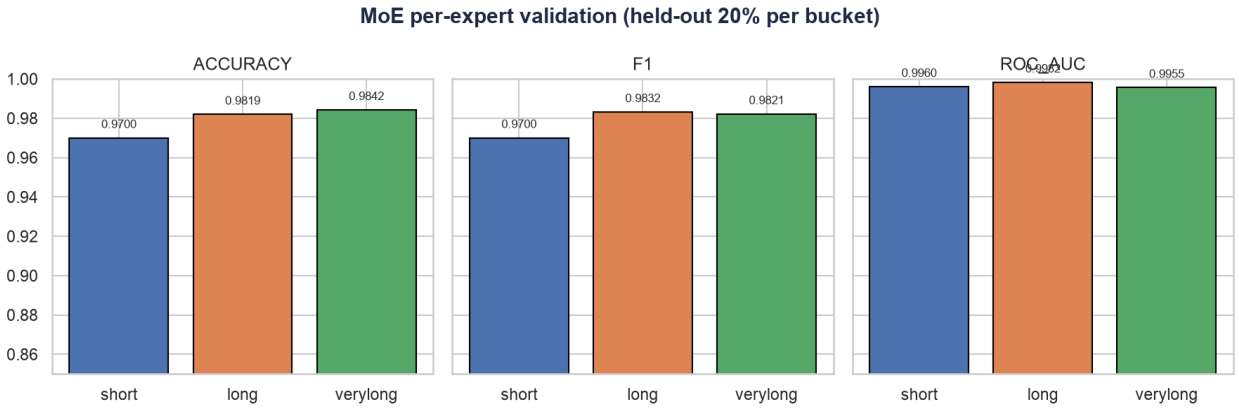


Figure A4: Per-expert accuracy/F1/AUC (all ≥0.97).

E Routed MoE vs Baseline (5k held-out + new domains)

Routing = length → expert. Full-text MoE costs 0.3pp vs baseline (98.96% vs 99.26%) — the price of specialization. Everywhere else it wins, and on unseen domains the gap is decisive (HC3: 48.6%→98.5%, modern: 85.5%→95.7%).

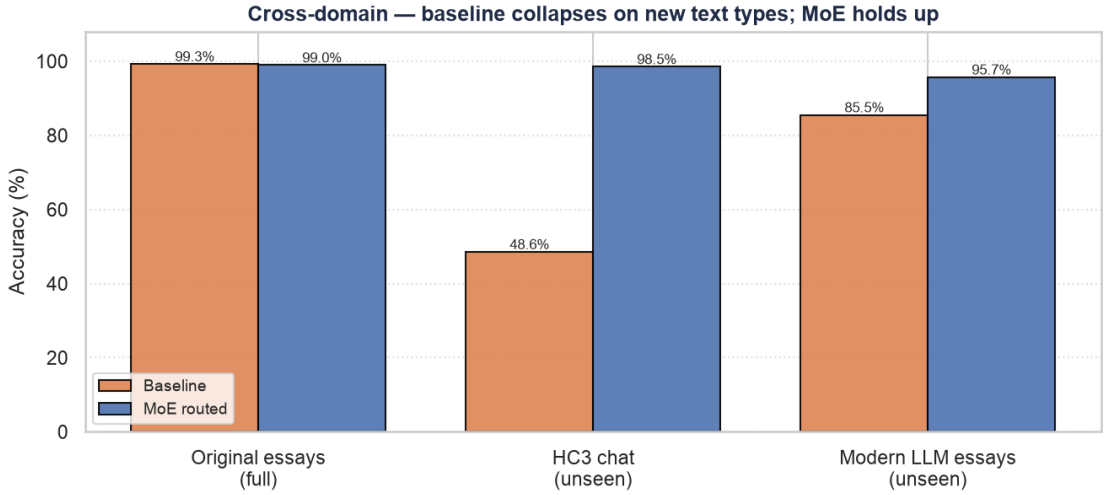


Figure A5: Cross-domain held-outs. Baseline is worse than chance on HC3 chat.

Condition	Baseline acc	MoE acc	Delta
full	99.26%	98.96%	+-0.3pp
first 25w	67.42%	86.06%	+18.6pp
first 50w	83.28%	93.12%	+9.8pp
first 1 sent	59.82%	81.64%	+21.8pp
HC3 holdout	48.56%	98.52%	+50.0pp
modern holdout	85.52%	95.70%	+10.2pp

Table A3: Headline comparison (results/moe_routed_metrics.csv + moe_baseline_comparison.csv).

Per-bucket on full text: verylong 99.59% (n=1,958), long 98.81% (n=2,945), short 90.72% (n=97 naturally-short essays).

F Threshold Calibration

The short expert under-calls AI on ultra-short essays (recall 0.68 at 25w). Calibrating on a calib half (2,500, seed 999, max F1) gave per-length thresholds 25w→0.32, 50w→0.40, 100w→0.56, 150w→0.64, implemented as `th=0.28+0.0024·n` clamped [0.32,0.64] for the short bucket (`models/moe/thresholds.json`). Long/verylong stayed at 0.50.

Eval-2500 condition	th=0.50	Adaptive	Note
first 25w	86.16% / F1 0.785	88.08% / F1 0.837	Recall 0.68→0.82
first 50w	93.12% / 0.905	93.08% / 0.909	Neutral (R↑ P↓)
first 100w	94.52% / 0.928	94.52% / 0.928	th≈0.52, identical
first 150w	94.96% / 0.935	96.20% / 0.949	+1.2pp
full	98.96%	98.92%	Flat

Table A4: Calibration on unseen eval half (results/moe_calibrated_metrics.csv). Caveat: HC3 natural short –0.6pp (human-heavy); essays gain. The <50w warning is retained.

Demo flips: a 31w borderline excerpt (P(AI)=0.374, th=0.354) is now AI-Generated (was Human under 0.50) — the intended fix, still low-confidence and flagged. A 14w casual human (P(AI)=0.035) stays clearly human.

G Discussion & Reproducibility

- **What MoE fixes:** length specialization + genuine short data. HC3 recall was the unlock — the baseline had never seen chat Q&A.; The 20k→16k per-expert vocabulary (vs 75k joint) also avoids long-essay features drowning short text.

- **What remains hard:** <25-50w is intrinsically weak (even MoE 86-93% with warnings). 150w essays still prefer the baseline 97.4% vs 95.6% — the short expert leans chat; blending experts at the boundary is future work.
- **Thresholds transfer:** calibration helps essay truncations but slightly hurts human-heavy HC3 natural short — thresholds are domain-sensitive; keep them length-adaptive, not global.
- **Type gate:** 80.5% is enough for routing insight, not for enforcement. Casual↔chat confusion is linguistic, not a bug.

Reproduce: `PYTHONPATH=. python3 moe/download_small_text.py` → `moe/train_moe.py` → `moe/calibrate_thresholds.py` → `moe/make_moe_appendix.py`. All artifacts under `models/moe/`, metrics under `results/moe_*.csv`.

Appendix References

- Guo, B. et al. How Close is ChatGPT to Human Experts? HC3 / Hello-SimpleAI/HC3. (2023).
- artnitolog/llm-generated-texts (GPT-4 Turbo/Omni, Claude 3 Opus, Llama3, Command R+ essay prompts).
- Roy, R. et al. A Comprehensive Dataset for Human vs AI Generated Text Detection (Defactify). arXiv:2510.22874 (2025) — context for cross-domain eval.
- Baseline report references apply (Hastie et al.; Robertson & Zaragoza; Sentence-BERT/DistilBERT/RoBERTa).